

Air University



Operating Systems (Lab)

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Ms. Fozia Riaz

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Department of Cyber Security

USAMA ARSHAD | 247141

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Lab Task # 02

“Linux Identity & File Detective”

Task Scenario

You are a new system user who has just logged into a Linux machine. Your goal is to identify yourself, explore the system, organize files, and investigate data using Linux commands.

Part A: Know Your System

- Display your username and system hostname

Command: whoami & hostname

```
(kali㉿kali)-[~]  
$ whoami  
kali  
  
(kali㉿kali)-[~]  
$ hostname  
kali
```

whoami displays the current user's ID, and hostname shows the system's network name. The result lists both on separate lines.

- Check how long the system has been running

Command: uptime

```
(kali㉿kali)-[~]  
$ uptime  
11:34:37 up 4 min, 1 user, load average: 0.06, 0.15, 0.08
```

This command shows how long the system has been active, along with load averages and user count. It indicates system stability and recent usage.

- Display full system information (kernel, OS, architecture)

Command: uname -a

```
(kali㉿kali)-[~]  
$ uname -a  
Linux kali 6.12.38+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.38-1kali1 (2025-08-12) x86_64 GNU/Linux
```

This displays comprehensive system information, including kernel version and hardware architecture. It provides a quick summary of the OS environment.

- Show the current date and calendar

Command: date && cal

```
(kali㉿kali)-[~]  
$ date && cal  
Fri Feb 13 11:40:53 AM EST 2026  
February 2026  
Su Mo Tu We Th Fr Sa  
1 2 3 4 5 6 7  
8 9 10 11 12 13 14  
15 16 17 18 19 20 21  
22 23 24 25 26 27 28
```

date outputs the current day and time, while cal displays the calendar for the current month. It helps verify system time settings.

- **Confirm your current working directory**

Command: pwd

```
(kali㉿kali)-[~]  
$ pwd  
/home/kali
```

This command prints the full pathname of the current working directory. It confirms exactly where you are in the file system.

Part B: Create Your Workspace

- **Create a directory named Lab2_<YourRollNo>**

Command: mkdir Lab2_247141

```
(kali㉿kali)-[~]  
$ mkdir Lab2_247141  
  
(kali㉿kali)-[~]  
$ ls  
Desktop Documents Downloads IA_Backup.tar Lab2_247141 Music Pictures
```

This creates a new directory named based on your roll number. It sets up a dedicated folder for your files.

- **Move inside this directory**

Command: cd Lab2_247141

```
(kali㉿kali)-[~]  
$ cd Lab2_247141
```

This command changes the current working directory to the specified folder. It moves you inside the new workspace.

- **Verify that you are in the correct directory**

Command: pwd

```
(kali㉿kali)-[~]  
$ cd Lab2_247141  
  
(kali㉿kali)-[~/Lab2_247141]  
$ pwd  
/home/kali/Lab2_247141
```

Used again to confirm that you have successfully moved into the Lab2_247141 directory. It ensures subsequent commands affect the correct folder.

- **Create three empty files:**

→ intro.txt

→ commands.txt

→ log.txt

Command: touch intro.txt commands.txt log.txt

```
(kali㉿kali)-[~/Lab2_247141]
$ touch intro.txt commands.txt log.txt

(kali㉿kali)-[~/Lab2_247141]
$ ls
commands.txt  intro.txt  log.txt
```

This command creates three empty files with the specified names. It prepares the files for content entry.

Part C: File Content & Investigation

- **Write your name, roll number, and semester into intro.txt**

Command: `echo -e "\nName:\tUsama Arshad\nRoll No:\t247141\nSemester:4th" > intro.txt`

```
(kali㉿kali)-[~/Lab2_247141]
$ echo -e "\nName:\tUsama Arshad\nRoll No:\t247141\nSemester:\t4th" > intro.txt
```

This command writes the provided text string directly into the specified file. It overwrites any existing content in the file.

- **Display the contents of intro.txt**

Command: `cat intro.txt`

```
(kali㉿kali)-[~/Lab2_247141]
$ cat intro.txt

Name:      Usama Arshad
Roll No:   247141
Semester:  4th
```

This displays the entire content of intro.txt on the terminal screen. It allows you to verify the file's data quickly.

- **Copy intro.txt into commands.txt**

Command: `cp intro.txt commands.txt`

```
(kali㉿kali)-[~/Lab2_247141]
$ cp intro.txt commands.txt

(kali㉿kali)-[~/Lab2_247141]
$ cat commands.txt

Name:      Usama Arshad
Roll No:   247141
Semester:  4th
```

This copies the content of intro.txt into a new file named commands.txt. It creates a duplicate file with the same content.

- **Count:**

→ Number of lines

→ Number of words

→ Number of characters in intro.txt

Command: `wc -l -w -c intro.txt`

```
(kali㉿kali)-[~/Lab2_247141]
$ wc -l -w -c intro.txt
4  8 51 intro.txt
```

This command counts and displays the number of lines, words, and characters in the file. It provides quick statistical analysis of the file data.

Part D: Smart Viewing & Sorting

- View only the first lines of commands.txt

Command: head commands.txt

```
(kali㉿kali)-[~/Lab2_247141]
$ head commands.txt

Name:      Usama  Arshad
Roll No:   247141
Semester:  4th
```

Displays the first ten lines of the file, allowing a quick preview of its content. It is useful for checking file headers.

- View only the last lines of commands.txt

Command: tail commands.txt

```
(kali㉿kali)-[~/Lab2_247141]
$ tail commands.txt

Name:      Usama  Arshad
Roll No:   247141
Semester:  4th
```

Displays the last ten lines of the file, useful for viewing recent log entries. It helps in inspecting the end of a file.

- Sort the contents of commands.txt

Command: sort commands.txt

```
(kali㉿kali)-[~/Lab2_247141]
$ sort commands.txt

Name:      Usama  Arshad
Roll No:   247141
Semester:  4th
```

This command sorts the lines of the file alphabetically and displays them. It organizes the data without modifying the original file.

Part E: Searching Like a Detective

- Search your name inside intro.txt

Command: grep usama intro.txt

```
(kali㉿kali)-[~/Lab2_247141]
$ grep usama intro.txt
```

Searches for the specific string "Name" within intro.txt and prints the matching lines. It is case-sensitive by default.

- **Perform a case-insensitive search**

Command: `grep -i Usama intro.txt`

```
(kali@kali)-[~/Lab2_247141]
$ grep -i Usama intro.txt
Name:      Usama Arshad
```

Performs a case-insensitive search for "name" and finds matches regardless of capitalization. It ensures all variations are located.

Part F: File & Directory Management

- **Rename commands.txt to system_info.txt**

Command: `mv commands.txt system_info.txt`

```
(kali@kali)-[~/Lab2_247141]
$ mv commands.txt system_info.txt

(kali@kali)-[~/Lab2_247141]
$ ls
intro.txt  log.txt  system_info.txt
```

Renames the file `commands.txt` to `system_info.txt`. It effectively updates the file's name in the directory.

- **Copy system_info.txt to your home directory**

Command: `cp system_info.txt ~/`

```
(kali@kali)-[~/Lab2_247141]
$ cp system_info.txt ~

(kali@kali)-[~/Lab2_247141]
$ cd

(kali@kali)-[~]
$ ls
Desktop  Documents  Downloads  IA_Backup.tar  Lab2_247141  Music  Pictures  Public  secret.enc  secret.enc~  secret.txt  system_info.txt
```

Copies `system_info.txt` from the current directory to your home directory. It creates a backup of the file in another location.

- **Delete log.txt**

Command: `rm log.txt`

```
(kali@kali)-[~/Lab2_247141]
$ ls
intro.txt  log.txt  system_info.txt

(kali@kali)-[~/Lab2_247141]
$ rm log.txt
```

This command deletes the file `log.txt` permanently. It is a dangerous command as files are not moved to a trash bin.

- **Verify deletion using listing commands**

Command: `ls`

```
(kali@kali)-[~/Lab2_247141]
$ ls
intro.txt  log.txt  system_info.txt

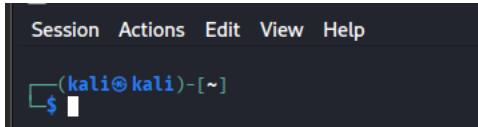
(kali@kali)-[~/Lab2_247141]
$ rm log.txt

(kali@kali)-[~/Lab2_247141]
$ ls
intro.txt  system_info.txt
```

Lists all files and directories in the current folder. It confirms that log.txt is gone and system_info.txt remains.

- **Clear the terminal screen**

Command: clear



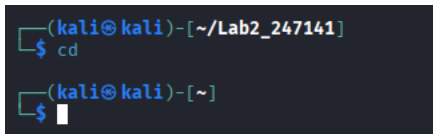
```
Session Actions Edit View Help
(kali@kali)-[~]
$ clear
```

Clears all previous text from the terminal screen. It provides a clean workspace for the next set of commands.

Part G: Final Cleanup

- **Move back to your home directory**

Command: cd

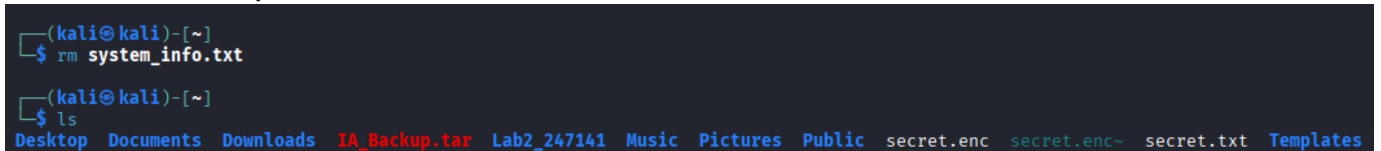


```
(kali@kali)-[~/Lab2_247141]
$ cd
(kali@kali)-[~]
$
```

Change the directory to your home folder. It brings you out of the lab directory.

- **Remove the copied file from home**

Command: rm system_info.txt

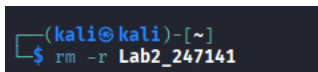


```
(kali@kali)-[~]
$ rm system_info.txt
(kali@kali)-[~]
$ ls
Desktop  Documents  Downloads  IA_Backup.tar  Lab2_247141  Music  Pictures  Public  secret.enc  secret.enc~  secret.txt  Templates
```

Deletes the file previously copied to the home directory. It removes the backup file.

- **Remove the Lab2_<YourRollNo> directory completely**

Command: rm -r Lab2_247141

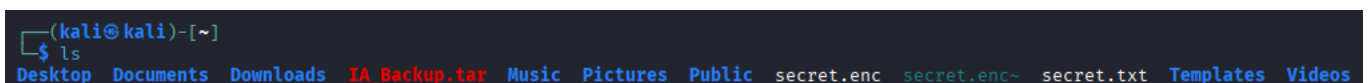


```
(kali@kali)-[~]
$ rm -r Lab2_247141
```

Recursively removes the Lab2_12345 directory and all its contents. It ensures the complete deletion of the workspace.

- **Verify that everything is deleted successfully**

Command: ls

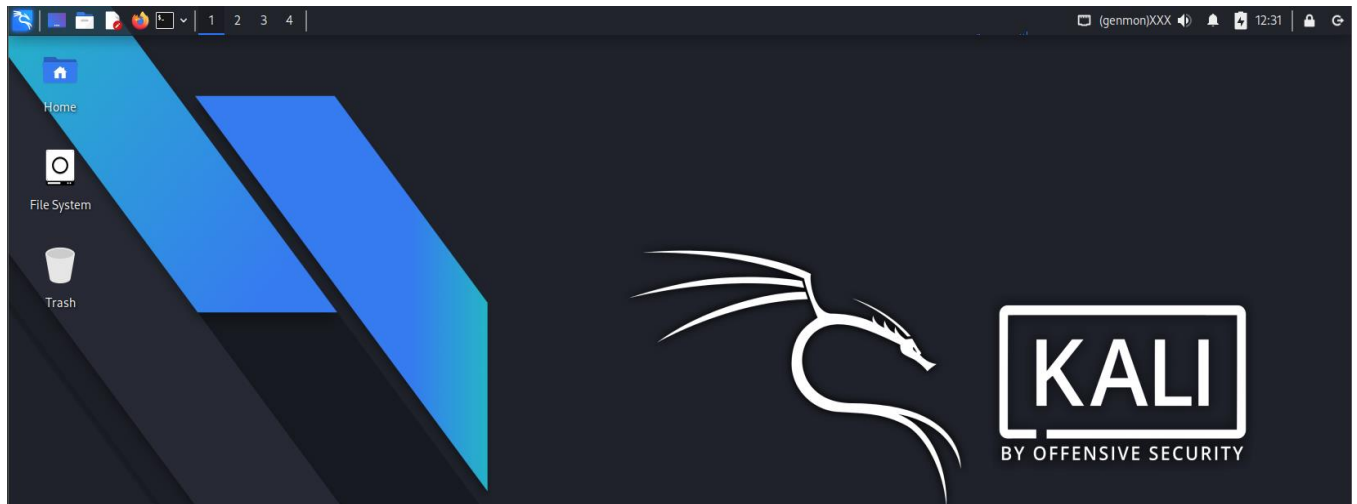


```
(kali@kali)-[~]
$ ls
Desktop  Documents  Downloads  IA_Backup.tar  Music  Pictures  Public  secret.enc  secret.enc~  secret.txt  Templates  Videos
```

Attempts to list the specific directory to verify it no longer exists. It confirms the deletion was successful.

- **Exit the shell**

Command: exit



Closes the current terminal session. It terminates the shell process.

THE END
